

Exercise 01:

1. Print numbers from 1 to 10

Using WHILE

```
i ← 1;

WHILE i ≤ 10 DO:

    PRINT i;

    i ← i + 1;

END WHILE
```

Using REPEAT...UNTIL

```
i ← 1;

REPEAT:

    PRINT i;

    i ← i + 1;

UNTIL i > 10;
```

2. Print the first 10 even numbers

WHILE

```
i ← 1;

WHILE i ≤ 10 DO:

    PRINT 2 * i;
```

```
    i ← i + 1;  
  
END WHILE
```

REPEAT...UNTIL

```
i ← 1;  
  
REPEAT:  
  
    PRINT 2 * i;  
  
    i ← i + 1;  
  
UNTIL i > 10;
```

3. Print the first n even numbers

```
READ n;  
  
i ← 1;  
  
WHILE i ≤ n DO:  
  
    PRINT 2 * i;  
  
    i ← i + 1;  
  
END WHILE
```

4. Print the next 10 even numbers starting from n (n is even)

```
READ n;  
  
i ← 1;  
  
WHILE i ≤ 10 DO:  
  
    PRINT n + 2 * i;  
  
    i ← i + 1;
```

```
END WHILE
```

5. Print the first 10 multiples of a given number n

```
READ n;  
  
i ← 1;  
  
WHILE i ≤ 10 DO:  
  
    PRINT n * i;  
  
    i ← i + 1;  
  
END WHILE
```

6. Compute and display the sum of numbers from 1 to 10

```
sum ← 0;  
  
i ← 1;  
  
WHILE i ≤ 10 DO:  
  
    sum ← sum + i;  
  
    i ← i + 1;  
  
END WHILE  
  
PRINT sum;
```

7. Compute the product of all integers between n and k (including n and k)

```
READ n, k;  
  
product ← 1;  
  
i ← n;  
  
WHILE i ≤ k DO:
```

```
product ← product * i;  
  
i ← i + 1;  
  
END WHILE  
  
PRINT product;
```

8. Compute the sum of all odd integers between n and k

```
READ n, k;  
  
sum ← 0;  
  
i ← n;  
  
WHILE i ≤ k DO:  
  
    IF i mod 2 < > 0 THEN:  
  
        sum ← sum + i;  
  
    END IF;  
  
    i ← i + 1;  
  
END WHILE;  
  
PRINT sum;
```

9. Compute the sum of several integers entered by the user (stop when 0)

Using WHILE

```
sum ← 0;  
  
READ x;  
  
WHILE x < > 0 DO:  
  
    sum ← sum + x;
```

```
    READ x;  
  
END WHILE;  
  
PRINT sum;
```

Using REPEAT...UNTIL

```
sum  $\leftarrow$  0;  
  
REPEAT:  
  
    READ x;  
  
    sum  $\leftarrow$  sum + x;  
  
UNTIL x = 0;  
  
PRINT sum;
```

Exercise 02: Factorial

```
READ n;  
  
factorial  $\leftarrow$  1;  
  
i  $\leftarrow$  1;  
  
WHILE i  $\leq$  n DO:  
  
    factorial  $\leftarrow$  factorial * i;  
  
    i  $\leftarrow$  i + 1;  
  
END WHILE;  
  
PRINT "n! =", factorial;
```

Exercise 03: Quotient & Remainder, Product, Power

a) Quotient and remainder without / operator

```
READ A, B;

quotient ← 0;

remainder ← A;

WHILE remainder >= B DO:

    remainder ← remainder - B;

    quotient ← quotient + 1;

END WHILE;

PRINT "Quotient =", quotient;

PRINT "Remainder =", remainder;
```

b) Product using successive additions

```
READ a, b;

product ← 0;

i ← 1;

WHILE i <= b DO:

    product ← product + a;

    i ← i + 1;

END WHILE;

PRINT "Product =", product;
```

c) Power using successive multiplications

```
READ a, b;
```

```
power ← 1;

i ← 1;

WHILE i ≤ b DO:

    power ← power * a;

    i ← i + 1;

END WHILE;

PRINT "a^b =", power;
```

Exercise 04: Sum and Average of numbers between two limits

```
READ lower, upper;

sum ← 0;

count ← 0;

i ← lower;

WHILE i ≤ upper DO:

    sum ← sum + i;

    count ← count + 1;

    i ← i + 1;

END WHILE;

average ← sum / count;

PRINT "Sum =", sum;

PRINT "Average =", average;
```

Exercise 05: Count occurrences of a character in a string

```
READ string;

READ char_to_count;

count ← 0;

i ← 1;

WHILE i <= LENGTH(string) DO:

    IF string[i] = char_to_count THEN:

        count ← count + 1;

    END IF;

    i ← i + 1;

END WHILE;

PRINT "Occurrences =", count;
```

Exercise 06: Check if a string is a palindrome

```
READ string;

length ← LENGTH(string);

is_palindrome ← TRUE;

i ← 1;

WHILE i <= length / 2 DO:

    IF string[i] <> string[length - i + 1] THEN:

        is_palindrome ← FALSE;

        BREAK;

    END IF;
```



```

    i ← i + 1;

END WHILE;

IF is_palindrome THEN:

    PRINT "Palindrome";

ELSE

    PRINT "Not a palindrome";

END IF

```

Exercise 07: Greatest power of $2 \leq n$

```

READ n;

power ← 1;

WHILE power * 2 ≤ n DO:

    power ← power * 2;

END WHILE;

PRINT "Greatest power of  $2 \leq n$ :", power;

```

Exercise 08: Divisors and odd divisors of an integer

a) All divisors

```

READ n;

i ← 1;

WHILE i ≤ n DO:

    IF n mod i == 0 THEN:

        PRINT i;
    
```

```
END IF;

i ← i + 1;

END WHILE;
```

b) Odd divisors

```
READ n;

i ← 1;

WHILE i ≤ n DO:

    IF n mod i == 0 AND i mod 2 < > 0 THEN:

        PRINT i;

    END IF;

    i ← i + 1;

END WHILE;
```

Exercise 09: Check if an integer is prime

```
READ A;

IF A ≤ 1 THEN:

    PRINT "Not prime";

ELSE

    is_prime ← TRUE;

    i ← 2;

    WHILE i ≤ A / 2 DO:

        IF A mod i == 0 THEN:
```

```
    is_prime  $\leftarrow$  FALSE;  
  
    BREAK;  
  
END IF;  
  
i  $\leftarrow$  i + 1;  
  
END WHILE;  
  
IF is_prime THEN:  
    PRINT "Prime";  
  
ELSE  
    PRINT "Not prime";  
  
END IF;  
END IF;
```